

DecodeLabs – Data Analytics Project 2

Exploratory Data Analysis (EDA)

Dataset: Dataset 2.xlsx

This report follows the supplied Project 2 requirements: calculate basic statistics (mean, median, count), identify trends and outliers, and summarize key observations.

Dataset size: 1200 rows × 14 columns

Numeric columns: Quantity, UnitPrice, ItemsInCart, TotalPrice

1. Basic Statistics

Column	Count	Mean	Median	Min	Max
Quantity	1200	2.95	3.00	1.00	5.00
UnitPrice	1200	356.41	364.21	11.39	699.93
ItemsInCart	1200	5.49	5.00	1.00	10.00
TotalPrice	1200	1,053.97	823.62	11.39	3,456.40

2. Outlier Analysis (IQR Method)

Column	Q1	Q3	IQR	Lower	Upper	Outliers
Quantity	2.00	4.00	2.00	-1.00	7.00	0
UnitPrice	186.06	521.57	335.51	-317.20	1,024.83	0
ItemsInCart	4.00	7.00	3.00	-0.50	11.50	0
TotalPrice	410.52	1,578.47	1,167.95	-1,341.41	3,330.41	8

An observation is flagged as an outlier when it is below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

3. Category / Trend Summaries

OrderID

Value	Count
ORD200000	1
ORD200001	1
ORD200002	1
ORD200003	1
ORD200004	1
ORD200005	1
ORD200006	1
ORD200007	1
ORD200008	1
ORD200009	1

Date

Value	Count
2023-08-20 00:00:00	8
2024-04-07 00:00:00	7
2024-12-19 00:00:00	6
2025-03-15 00:00:00	6
2024-11-24 00:00:00	6
2023-02-16 00:00:00	5
2024-07-24 00:00:00	5
2024-08-31 00:00:00	5
2024-06-08 00:00:00	5
2024-07-22 00:00:00	5

CustomerID

Value	Count
C14847	2
C98474	2
C21191	2
C46651	2
C97593	2
C56969	2
C35852	2
C38840	2
C94569	2
C70659	2

Product

Value	Count
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170
Monitor	163
Phone	156

ShippingAddress

Value	Count
533 Main St	8
895 Main St	6
562 Main St	5
146 Main St	5
740 Main St	5
696 Main St	5
636 Main St	5
747 Main St	5
904 Main St	4
128 Main St	4

PaymentMethod

Value	Count
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

OrderStatus

Value	Count
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

TrackingNumber

Value	Count
TRK37947903	1
TRK91186779	1
TRK42903982	1
TRK62788070	1
TRK29241424	1
TRK72976927	1
TRK96417362	1
TRK78809193	1
TRK61042692	1
TRK33478363	1

CouponCode

Value	Count
FREESHIP	313
nan	309
WINTER15	292
SAVE10	286

ReferralSource

Value	Count
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

4. Key Observations

- The dataset contains 1200 records across 14 columns.
- Quantity has a mean of 2.95 and a median of 3.00.
- UnitPrice has a mean of 356.41 and a median of 364.21.
- ItemsInCart has a mean of 5.49 and a median of 5.00.
- TotalPrice has a mean of 1,053.97 and a median of 823.62.
- Quantity has 0 IQR-flagged outlier(s).
- UnitPrice has 0 IQR-flagged outlier(s).
- ItemsInCart has 0 IQR-flagged outlier(s).
- TotalPrice has 8 IQR-flagged outlier(s).
- The most frequent value in OrderID is 'ORD200000' with 1 record(s).
- The most frequent value in Date is '2023-08-20 00:00:00' with 8 record(s).

- The most frequent value in CustomerID is 'C14847' with 2 record(s).
- The most frequent value in Product is 'Printer' with 181 record(s).
- The most frequent value in ShippingAddress is '533 Main St' with 8 record(s).
- The most frequent value in PaymentMethod is 'Online' with 258 record(s).
- The most frequent value in OrderStatus is 'Cancelled' with 250 record(s).
- The most frequent value in TrackingNumber is 'TRK37947903' with 1 record(s).
- The most frequent value in CouponCode is 'FREESHIP' with 313 record(s).
- The most frequent value in ReferralSource is 'Instagram' with 259 record(s).

5. Conclusion

The EDA summarizes the dataset using count, mean, and median, checks numeric variables for IQR-based outliers, and identifies the most frequent categorical values. These findings provide a descriptive view of the dataset before further analysis or modeling.